

Experimental Study on the Use of Rocks and Minerals in the Development of High-Temperature Ceramic Glazes

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Ceramic glazing represent one of the most fertile meeting grounds between art, craftsmanship, and science. It brings together practical and theoretical knowledge, aesthetic sensibility, and complex chemical-physical processes. This article is drawn from a thesis that arose from the desire to explore new possibilities in the formulation of high-temperature ceramic glazes, shifting the focus away from conventional industrial materials towards natural minerals and rocks. The raw minerals used - such as aquamarine, chrysocolla, diopside, obsidian, and lapis lazuli - were tested in combination with fluxes, stabilizers, and other functional oxides to evaluate their behaviour during firing, chromatic response, surface texture, and compatibility with different ceramic bodies.

In the Ancient Near East, Egypt, and China, the historical evolution of ceramic glazing highlights the crucial role of mineralogical knowledge in shaping artistic and symbolic practices.

In Egyptian and Mesopotamian cultures, steatite was coated with vitreous mixture composed of silica, alkaline fluxes (such as natron), and copper-based pigments, resulting in blue-green surfaces symbolically associated with the divine realm (Alwardany, 2023).



Fig. 1 Ovoid scarab in steatite coated with a glossy blue-green glaze. Ancient Egypt, Third Intermediate Period, Dynasties XXI - XXV, ca. 1069-664 B.C.E. Africart Market.

Fig. 2 Bowl with fish and lotus flowers. Western Thebes, New Kingdom of Egypt, ca. 1450-1400 B.C.E. The Walters Art Museum.

At the same time, Egyptian faience, composed of quartz, lime, and plant ash, employed copper as its primary colorant, while in Mesopotamia experiments were conducted with the addition of alumina, potash, and iron oxides to improve glaze adhesion.



Fig. 3 Vase depicting a dragon. China, Jiangxi Province, Ming Dynasty, 1426-1435. The Met Museum.

Fig. 4 Earthenware bowl with polychrome underglaze decoration. Iran, Nishapur, late 8th early 9th century. Wikipedia



In China, technical excellence was grounded in the careful use of minerals such as cobalt oxide and carbonate for blue tones, malachite and chrysocolla for greens and reds, hematite and magnetite for iron-based variations, and pyrolusite for blacks and purples - achieving a sophisticated control over the reactions occurring in oxidizing or reducing atmospheres.

In the Islamic world, the ceramic production of Nishapur made use of quartz, malachite, cassiterite, calcite, and cobalt compounds to formulate alkaline glazes on siliceous bodies. In the centers of Kashan and Rayy, techniques were further refined through the use of galena, natron, feldspars, and chromite, culminating in metallic luster glazes based on copper and silver.

In Anatolia, Seljuk ceramics incorporated quartz sand, lead oxide derived from galena, cassiterite, kaolin, wollastonite, and at times chromite to regulate viscosity and the thermal response of glazes - demonstrating a profound empirical understanding of materials and a refined synthesis of science, aesthetics, and symbolism.



Fig. 5 Ciotola a lustro, stonepaste, smaltata con sottosmalti blu e turchese. Siria, Raqqa, fine XII prima metà XIII secolo. The Met Museum.

Minerals and Rocks Selected for the Research

In selecting the minerals used in this study, consideration was given not only to their chemical composition, but also to several fundamental physical properties such as hardness, tenacity, density, and lustre. After careful evaluation, the minerals and rock samples selected were as follows:

aquamarine ($\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$), obsidian (rock), chrysocolla ($\text{Cu}_{2-x}\text{Al}(\text{H}_{2-x}\text{Si}_2\text{O}_5)(\text{OH})_4 \cdot n\text{H}_2\text{O}$, $x < 1$), diopside ($\text{CaMgSi}_2\text{O}_6$), and lapis lazuli ($\text{Na}_7\text{Ca}(\text{Al}_6\text{Si}_6\text{O}_{24})(\text{SO}_4)(\text{S}_3) \cdot \text{H}_2\text{O}$).

A key criteria in the final selection was the presence of a significant percentage of silica (SiO_2) in the composition of each mineral (and rock), an element of primary importance in the formulation and melting processes of ceramic glazes.

The Development of the Formulas

At the foundation of any glaze composition lies silica (SiO_2), a vitreous oxide that serves as the primary glaze forming agent. However, its high melting point (1700°C) makes it unsuitable for direct use in ceramic firing, which occurs at lower temperatures. For this reason, fluxin oxides - such as those of lead, sodium, potassium, calcium, magnesium, barium, zinc, and boron - are added to lower the melting temperature of the glaze.

Stabilizing agents, particularly alumina (Al_2O_3), regulate the viscosity of the molten mass, improving both its mechanical strength and adhesion to the ceramic body. The choice and balance of these oxides, depending on the firing type and temperature, ultimately determine the aesthetic and technical quality of the glaze.

The raw materials - including feldspar, kaoli, dolomite, wollastonite, and quartz - supply the required oxides in the appropriate proportions, while colouring oxides (such as chromium, copper, cobalt, iron, manganese, and tin oxides) provide the desired chromatic range according to the firing atmosphere.

The calculation of glaze formulas is based on a molecular balance among the three functional groups - glass formers, fluxes, and stabilizers - which ensure glazes with consistent melting behaviour, adhesion, and color stability. This chemical-experimental approach allows for a degree of control and reproducibility in the final results.

GLAZE "LIMIT" FORMULAS

	TYPE OF GLAZE	CONE RANGE	LIMIT FORMULA						
1125°-1180°C	Fritted whiteware transparent lead borosilicate	1 - 5	PbO KNaO CaO ZnO	2-3 2-3 .35-.5 0-.1	Al_2O_3 B_2O_3	.25-.35 .3-.7	SiO_2	2.5-3.5	
1165°-1225°C	Bristol stoneware	4 - 8	KNaO CaO MgO ZnO	.25-.4 .1-.3 0-.2 .15-.4	Al_2O_3	.35-.55	SiO_2	2.5-4.0	
1125°-1300°C	Stoneware or porcelain	8 - 12	KNaO CaO MgO ZnO	2-4 4-7 0-3 0-2	Al_2O_3	.3-.5	SiO_2	3.0-5.0	

Table 1. Limits for the calculation of glaze formulas.

The Tests

For the purposes of the experimental research, two samples formats were produced: small bowls made of bisque porcelain and tiles in four different refractory clays, in order to analyze the behaviour of the glazes on surfaces of differing nature and colour. The bowls were used to test glaze formulations, while the tiles, in their varying shader, allowed for the evaluation of coverage, distribution, and chromatic variations of the glazes applied by brush, as well as the observation of colour reactions in relation to different substrates.

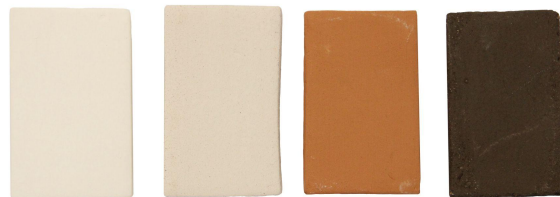


Fig. 6 The four types of refractory clays: porcelain, white semi-refractory, red refractory, and black semi-refractory.

The minerals employed, originating from various geographical areas, were crushed, ground, and sieved until obtaining fine, homogeneous powders. Their chemical composition was then determined through a calculation of molecular equivalents, a method that allows one to derive the simplified formula of each mineral based on the weight percentages of its constituent elements.

The data, obtained from geochemical sources (Mineralogy Database), were converted into moles by dividing the percentages by their respective molecular weights, and subsequently normalized to the oxygen content. This procedure made it possible to express the molecular composition of each sample in proportional form - for example, for aquamarine, the resulting formula was $2\text{SiO}_2 \cdot 0.34\text{Al}_2\text{O}_3 \cdot \text{BeO}$, allowing for a precise definition of the mineral's chemical contribution within the glaze composition.

Once the recipe of each glaze had been defined and converted into weight percentages, the raw materials, minerals, and rocks were carefully weighted. Each formulation was prepared with a dry weight of 10 g for the bowls and 5 g for the tiles, with the optional addition of colouring oxides in percentages ranging from 1% to 12%. To each sample, 10 g of water was added, then mixed and filtered until a homogeneous, impurity-free suspension was obtained. The mixture was applied to the supports, left to dry, and fired in an electric kiln at 1240°C in an oxidizing atmosphere.



The Evaluation of the Results

The analysis focused on technical parameters such as fluidity, adhesion, coverage, surface uniformity, and color stability, as well as on the effects of colouring oxides. Each mineral was tested in different formulations to identify the most stable and effective combinations, resulting in a total of 150 samples, including both bowls and tiles.

Aquamarine-based glazes exhibited good adhesion and consistent melting behaviour, producing neutral tones that, through the addition of cobalt, chromium, and iron oxides, yielded chromatic variations ranging from deep blue to sage green to warm brown, though occasionally showing microbubbles caused by fusion gases.



Fig. 7-8-9 Aquamarine. Samples with cobalt, chromium, and red iron oxide, from left to right.

Chrysocolla glazes, characterized by the presence of CuO , generated glossy or metallic surfaces depending on the mineral concentration, achieving the most aesthetically refined results in low-copper formulations. The addition of coloring oxides produced stable and uniform hues, from midnight blue to rust brown.



Fig. 10-11 Chrysocolla. Two different formulations containing different percentages of chrysocolla - 17% on the left, 48% on the right.

In the case of diopside, varying the proportions of alkali and alkaline-earth fluxes made it possible to obtain stable, opaque, and defect-free glazes, with gradual transitions in finish from satin to glossy. The addition of oxides created well-balanced color variations, from sapphire blue to reddish brown, without structural alterations.

Obsidian-based glazes showed a high level of compatibility with glassy matrices, although variations in alumina and flux content affected viscosity and brilliance. The final formulations, containing reduced quantities of raw materials, produced compact, defect-free surfaces with warm, uniform tones.



Fig. 12-13-14-15-16 Diopside. Top left: diopside and clear high-temperature glaze, top right: recipe 4. Bottom row, from left to right: formulations with cobalt, chromium, and red iron oxide.



Fig. 17-18-19-20-21 Obsidian. Top left: pure obsidian sample; right: recipe 7. Bottom row, from left to right: additions of cobalt, chromium, and red iron oxide.

Finally, tests with lapis lazuli revealed the loss of its characteristic blue coloration due to the volatilization of lazurite above a certain temperature, yet resulted in mother-of-pearl-like, craquelé surfaces of notable aesthetic interest.



Fig. 22-23 Lapis lazuli. Left: with clear high-temperature glaze; right: pure.

Conclusions

This research has demonstrated that the use of natural minerals and rocks in the formulation of high-temperature ceramic glazes represents a field of inquiry capable of combining scientific rigor with aesthetic exploration.

Each mineral revealed distinct behaviors during firing, determined by its chemical composition and its interaction with the other components of the glaze. Aquamarine and chrysocolla displayed excellent chromatic and textural potential, while more complex minerals, such as lapis lazuli, revealed limitations linked to the volatilization of their constituent elements.

The addition of colouring oxides proved essential in expanding the tonal range and in controlling variations in brilliance and saturation.

From a technical perspective, the tests conducted made it possible to accurately evaluate fluidity, adhesion, stability, and potential defects, providing fundamental insights for defining balanced and reproducible formulations.

The results obtained also open up to the possibility of applying this knowledge in the artistic field, where the developed glazes can serve as expressive tools, capable of translating symbolic and poetic concepts through matter itself.

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